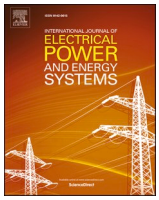
Electrical Power and Energy Systems 153 (2023) 109349



Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes





A bidirectional loss allocation method for active distributed network based on Virtual Contribution Theory

Jing Zhao [a], Songhuai Du [a], Yanjun Dong [a], Juan Su [a,*], Yue Xia [a]

[a] College of Information and Electrical Engineering, China Agricultural University, Beijing, China

ARTICLE INFO

Keywords:
Virtual contribution theory
Bidirectional power loss allocation
Power flow tracing
Active distribution network
Electricity market

ABSTRACT

Loss Allocation is a crucial task in power transmission and distribution services, as it essential for reconciliation of generator-user settlement results. With the availability of distributed generation in recent years, power can flow in both directions along transformers and mainlines in the traditional distribution networks. This makes it difficult to calculate power losses in active distribution networks and correct transactions between generators and users. In this paper, we propose the concept of Virtual Contribution Theory and construct a virtual contribution matrix to illustrating how generators and loads use and occupy distribution networks. By combining Virtual Contribution Theory with Power Flow Tracing, a bidirectional loss allocation method based on the Virtual Contribution Theory is proposed to account for power losses in active distribution network. One key advantage of this method is that it eliminates many shortcomings of traditional loss allocation method, including cumbersome calculation processes, low solution efficiency, and limitations in bidirectional power flow calculations. This method can also fairly allocate power loss between generators and loads in the retail electricity market. We conducted tests on IEEE 33 and IEEE 69 systems to evaluate the feasibility and effectiveness of the proposed method.

1. Introduction

Since the rapid development of distributed energy resources (DER) in recent years, a global consensus on climate change and energy strategy has emerged, leading countries to advocate for a transition towards low-carbon and sustainable energy systems. However, the intermittent nature and inherent uncertainties of DER present challenges, necessitating a shift from passive to active distribution networks(ADN) as large-scale DERs are connected to grids and the terminal load prosumer pattern emerges [1–4]. The proliferation of flexible DER is accelerating the development of the distributed energy market and driving up operational economy and technical management standards for ADNs. Economic agents in ADN are also continuously improving their self-awareness and demand management, leading to a heightened focus on economic management for reliable power supply [5,6]. Especially after the blackouts of 2003 [7] and 2020 [8], the aggregation effect on power loss for grid connections by independent individuals in complex ADNs has become increasingly evident. For instance, how does the decentralized DER's grid-connected behavior affect power losses in complex ADN? Furthermore, how can we quantify the contribution of each independent agent to the overall power loss in a complex power system?

Connecting distributed generation (DG) to the distribution network increases its flexibility. However, there is often a two-way flow between the transformer branch and trunk branch [9,10], which can result in difficulty when calculating the power loss in ADN and correcting the transaction results for power generators and users. Moreover, managing energy consumption and recognizing the system's operation become more complex. Traditional distribution energy management can no longer meet and adapt to the needs and the development of new power systems under the energy transition. This highlights the non-transparent and discriminatory nature of subject cognition. In addition, the experience of electricity markets in the United Kingdom suggests that loss allocation methods (LA) can influence energy transaction prices by up to 10% [11], raising questions about how LAs can coexist with energy management. The development of the methods to quantify the contributions of different entities to power losses is particularly relevant for analyzing and managing the power quality system. Therefore, efficient loss management and distribution measures are essential to maximizing economic benefits for all participants, achieving high efficiency, reducing energy consumption, and ensuring fairness.

Quantifying the contribution of individuals to the collective behavior

* Corresponding author.
E-mail address: sujuan@cau.edu.cn (J. Su).

https://doi.org/10.1016/j.ijepes.2023.109349
Received 18 December 2022; Received in revised form 9 June 2023; Accepted 25 June 2023
Available online 2 July 2023
0142-0615/© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Nomenclature

LA	Loss Allocation
(A)DN	(Active) Distribution Network
VCT	Virtual Contribution Theory
PSR	Proportionate Sharing Rule
SS	Secondary Substation
Y	Admittance Matrices
Z	Impedance Matrices
DER	Distributed Energy Resources
DG	Distributed Generation
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
MRE	Mean Relative Error

Indices

n, k	Number of nodes and branches
m, l	Number of generators and loads
P_{Gi}	Generator output for G_i at Node i
P_{Li}	Power demand for L_i at Node i
$i \rightarrow j$	Downstream direction from Node i to Node j
$j \rightarrow i$	Upstream direction from Node j to Node i
$P_{i \rightarrow j}$	Input power at Node j
$P_{j \rightarrow i}$	Output power at Node j
ρ_{jin}	The ratio of input power at Node j
ρ_{jout}	The ratio of output power at Node j
$P_{j,in}$	Input power for Branch j
$P_{j,out}$	Output power for Branch j
β	Shared coefficient for loss allocation

Vectors and matrices

$s_1[j]$	Vector of start Node in Branch j
$s_2[j]$	Vector of end Node in Branch j
VGM	$m \times n$ matrices where element $\mathbf{VGM_{ij}}$ is present the virtual generator G_i at Node j
VLM	$l \times n$ matrices where element $\mathbf{VLM_{ij}}$ is present the virtual load L_i at Node j
G −VCPM	$m \times n$ matrices where element $\mathbf{G - VCPM_{ij}}$ is present the virtual contribution ratio at Node j by G_i
L −VCPM	$l \times n$ matrices where element $\mathbf{L - VCPM_{ij}}$ is present the virtual contribution ratio at Node j by L_i
G −VOM	$m \times k$ matrices where element $\mathbf{G - VOM_{ij}}$ is present the virtual occupancy on Branch j by G_i
L −VOM	$l \times k$ matrices where element $\mathbf{L - VOM_{ij}}$ is present the virtual occupancy on Branch j by L_i
G −VOPM	$m \times k$ matrices where element $\mathbf{G - VOPM_{ij}}$ is present the virtual contribution proportion on Branch j by G_i
L −VOPM	$l \times k$ matrices where element $\mathbf{L - VOPM_{ij}}$ is present the virtual contribution proportion on Branch j by L_i
$\mathbf{P_{LOSS}}$	$k \times k$ matrices only have nonzero elements at diagonal positions and its value correspond the branch loss
$\mathbf{P_{LOSS_G}}$	$m \times k$ matrices where element in row i and column j is present the allocated loss on the generation-side
$\mathbf{P_{LOSS_L}}$	$l \times k$ matrices where element in row i and column j is present the allocated loss on the load-side

of the power network is a significant challenge. Power flow tracing (PFT) introduces a novel idea and direction for addressing the above problems by decomposing and summarizing the power of each component. The use of PFT has led to transparent and efficient operation of the distribution network, creating an optimal regulatory and economic environment [12–14]. Additionally, PFTs have the potential to analyze, validate, and model economic concepts in current power system, such as energy trading [15], P2P trading [5,16], and regional Blockchain transactions [17,18]. PFT has already been widely employed in transmission pricing [19], harmonic tracking [20–22], loss allocation [13,23,24], and carbon footprint accounting [25].

This paper provides a review and analysis of three widely used PFT methods, and evaluates their potential application prospects for power system operation.

The first PFT method reviewed is based on circuit theory, which shares power proportionally between each bus [26]. This proportionate sharing rule (PSR) assumption has been shown to be valid in cooperative game theory and information theory [27,28], and yields the optimal cost allocation regardless of the cost function used. This method can be categorized based on its calculation logic and form, including topological distribution [29], graph theory [30,31], node power distribution [32], generator domain [28] and correlation matrix extension [33]. In [12], a method based on generator-load pairing is used to allocate power losses. However, this method requires further calculations to determine the loss allocation for each participant. This approach does not satisfy the transparency and openness of information gathering by each party, and is not conducive to the refined management of the energy system. In [31], graph theory is used to solve PFT. However, this method cannot differentiate the contributions of each node in the loop, and iteration is required. One limitation of circuit theory-based PFTs is that they may not fully capture all participant information. This can result in some participants not having access to detailed data on their contribution to power loss. Such method only allows for obtaining the total amount assumed by each participant without considering its source or composition. This can make it difficult to ensure transparency and non-discrimination among participants.

The second type of method aims to account for source and load losses directly by using their admittance (Y) or impedance (Z) matrices along with their load current equations [13,34,35]. However, this type of analysis can cause a secondary loss that cannot be separated due to the decomposition and reorganization of current and power supply. As a result, it is challenging to ensure fair and non-discriminatory results for loss and cost sharing. To address this issue, a new approach was proposed in [36,13], where the constant power element was replaced by an admittance element. This modification allows for current decomposition and tracking while keeping active and reactive power coupled. However, this method is computationally complex due to the inverse operation required on admittance-like matrices. Furthermore, as suggested by the authors in [35], decomposing load voltage-related items into generator-related items can lead to singularity issues with the inverse matrix, making the solution process computationally intensive and requiring significant storage space. This type of method is computationally heavy, requiring complex matrix operations with the Y matrix, generating high computational loads and storage demands, making it unsuitable for practical engineering applications.

By improving the above two methods, a third type of method has been developed to enhance the calculation efficiency of the PFT method and reduce tracking time. Matrix-based PFT is used to quantify and track the reactive power of the system in [37], but it still requires complex matrix inversion operations, which makes it difficult to improve its computational efficiency. The applicability of this method for active power tracking needs to be further verified as reactive power generated by a selected power source may not follow the direction of active power in the specified branch. In [38], a new method of power flow is presented that specifies how each nodal generation and demand contributes to a particular line's power flow, which reduces the degree of freedom of the system and eliminates the need to calculate the downstream and upstream distribution matrix. Although the calculation time is reduced,

this method still has high computational complexity. Some studies, such as [39–41], have combined artificial intelligence and PFT methods to determine the power supply share for a given load. These methods then use neural networks and support vector machines to train on the contribution share data. While this method overcomes some of the issues associated with the first two types of PFT, such as incomplete information acquisition, complex calculations, and low efficiency, its accuracy depends heavily on the training process and hyperparameter selection. Moreover, only the load and power supply contribution to branch power loss can be quantified by using this method. Therefore, the robustness and applicability of this method still need further improvement and optimization on technical issues.

As the prevalence of DGs in the energy sector increases, their integration into the grid can alter power flow and network loss. Various relevant studies [42–45] have identified crucial technical issues in actual DN resulting from the emergence of DERs, including voltage rise, reverse power flow, unintentional islanding, and voltage unbalance. These technical problems further amplify the complexity and diversity of the actual DN. This is also a primary reason why transmission loss allocation methods cannot be directly applied to distribution networks. Therefore, it is essential for LA method to accurately determine each participants' contributions to power loss. A comprehensive review of LA methods considering the DG is presented in [42]. In [46], a three-step LA method is proposed that first allocates loss to generators (including DGs) and loads separately, and then normalizes the results if the distribution loss does not match the total loss. In [47], a branch-oriented LA method is introduced, which considers three load models and four DG types and does not required further normalization. This method uses a power summation algorithm to calculate power flow. However, it only examines the scenario of a single DG access, and its suitability for situations with multiple DGs connected simultaneously needs further investigation. In [48], a novel LA method based on branch current decomposition is proposed that take into account reactive power transactions by network users and network reconfiguration, and ensures fair allocation of loss incentives. However, as observed in [47,48], negative loss can occur for generators or loads. Negative loss allocations can be seen as cross-subsidies that lack economic justification. Power trading interactions between generators and loads on transmission lines cause losses, and discouraging large-scale DG grid connections with a reward and punishment mechanism based on negative loss allocation is not feasible. Thus, our proposed method does not permit negative loss allocation, in line with the approach advocated in [12].

Although the existing PFT method is capable of accurately reflecting the trend of the network power flow and distribution of power distribution, further improvement is necessary. This is reflected in two main aspects: 1) The complex and cumbersome calculation process, which involves calculating downstream/upstream distribution matrix, class admittance matrix, and inverse matrix, making it time-consuming and problematic. 2) The high computational redundancy of the apportionment principle, which repeatedly calculates the fixed market member's contribution to individual lines, rending it unsuitable for engineering applications.

To address these issues, a VCT-based ADN bidirectional LA method is proposed, which addresses inefficiencies in fine-tuning of ADN energy management, incomplete information gathering, and discrimination issues. A space–time complexity analysis confirms the simplicity and efficiency of the proposed method, which outperforms the traditional LA method in terms of calculation efficiency and speed. The new method uses a virtual contribution matrix instead of repeatedly solving the inverse matrix in the traditional LA step, reducing computational complexity and improving efficiency. Hence, this method exhibits both technological innovation and advancement, as well as good scalability, enabling its application to areas such as carbon emission responsibility and water resource allocation. Additionally, our proposed method quantifies the contribution of diverse economic entities to power loss in the electricity retail market. This not only ensures fairness and non-discrimination in meeting multi-subject allocation requirements but also yields significant economic benefits. Initially, our proposed method instigates a transformative shift from the traditional extensive loss management paradigm to a more refined schema, substantially enhancing the energy management efficiency of ADNs. Subsequently, our model transcends the mere quantification of each participant's contribution to network asset utilization, instead providing a precise demarcation of high-loss zones within the system, thereby facilitating targeted implementation of loss reduction measures. Moreover, this strategy proffers the potential for utility and power transmission and distribution companies to undertake effective equipment maintenance and upgrades, resulting in a decrease in operational costs and improved quality of service provision. Finally, our approach promotes a fair and efficient allocation of power losses among various economic entities, which in turn, encourages a just and transparent distribution of profits and losses within the power market, thereby safeguarding the economic benefits of all market entities. Therefore, this study is of great significance for promoting the power marketization process, improving the efficiency and fairness of power markets, and has the potential to contribute to the sustainable development of the energy industry.

This article is divided into four sections. The first section provides an introduction to the topic. The second section focuses on the VCT and evaluates the contributions of each generator and load to the branch losses. In the third section, we compare and discuss the accuracy of the proposed method, as well as its calculation efficiency and applicability, using IEEE 33, 69 standard systems. Finally, the last section presents the conclusion.

2. Proposed virtual contribution based bidirectional loss allocation method

This section aims to clarify the concept and principle of VCT and introduce a novel bidirectional LA method based on it. In response to the limitations of existing PFT methods, such as the need to invert the distribution matrix, high calculation redundancy, low operation efficiency, and incomplete data acquisition, this paper proposes a bidirectional LA method that employs VCT based on PSR [26]. The proposed method accurately allocates power loss by measuring generator and load contributions to each branch.

2.1. The rationale of virtual contribution theory

A schematic diagram of VCT is shown in Fig. 1. Each node in the system can be represented as a virtual combination of generators and loads, resulting in a "virtual" decoupling between each node's input and output. For example, for node N_i, its input and output power flow come respectively from the upstream generator or local generator and the downstream load or local load. In node N_1, input power from G_1, and output power comes from B_4 and B_1. The sum of $G_1^{'}$ and $G_1^{''}$ represents the virtual generators of G_1 at nodes N_4 and N_2, respectively. Similarly, $L_2^{'}$ and $L_2^{''}$ represent the virtual loads of L_2 at nodes N_4 and N_2. This introduced VCT allows for a "virtual" decoupling between load and generator while maintaining the same number of nodes in the original system. The proposed method maintains the original system's physical topology, and therefore does not entail any changes in computational complexity or cost.

2.2. Bidirectional apportion of network loss based on virtual contribution Theory

To address the challenges of complex computation and low efficiency in existing methods, this paper proposes a VCT-based approach that constructs a virtual contribution matrix to quantify the degree of contribution of each generator or load to the power flow of each node and branch. With this method, power generation and load side losses can

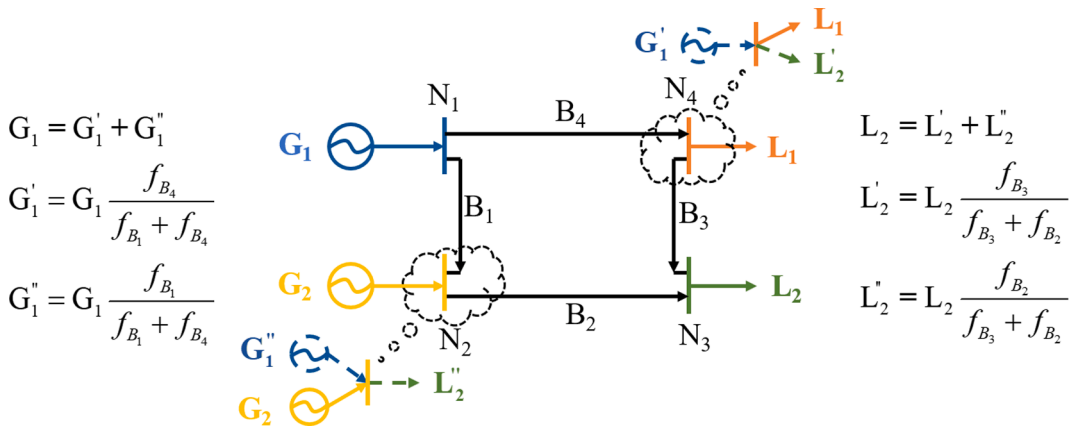


Fig. 1. Schematic diagram of Virtual Contribution Theory, where f_{B_i} represents the power flow on branch B_i.

be bidirectionally allocated. The proposed method is applied to a network with n nodes, k branches, m generators, and l loads, where G_i and L_i are located at the i_{th} node. The steps of this method are as follows:

2.2.1. Step 1: Build virtual generator matrix/ virtual load matrix VGM/VLM

In Fig. 1, each node in the system can be independently represented as a virtual combination of generators and loads that contribute to the node. A virtual generator/load matrix can then be constructed to quantify each generator's and load's contribution to each node. The first step is to construct the in-out node set of each node to determine the power transmitted direction or the source direction. For generators, the downstream tracking technique is used to decouple the generated power based on the input ratios of the nodes. The virtual generated power is then substituted for the input branch power corresponding to the outgoing node, creating the virtual generator matrix (**VGM**). Similarly, a virtual load matrix (**VLM**) is generated by decoupling the load power according to the output ratio of a node. This is done by replacing the output branch power corresponding to the input node with the virtual load power.

Generators are given priority to supply local loads, with any excess power being distributed proportionally. Using established power flow calculations, the **VGM** and **VLM** are calculated through downstream and upstream tracing, respectively. The calculation formula is as follows:

$$\mathbf{VGM} = (\mathbf{VGM_{ij}})_{\mathbf{m\times n}} = \begin{cases} P_{Gi} i = j, i \in [1,m], j \in [1,n] \\ \sum_{S_{i\to j}} \prod_{N_{i\to j}} \rho_{jin} \times P_{i\to j} i \neq j, i \in [1,m], j \in [1,n] \end{cases} \quad (1)$$

$$\mathbf{VLM} = (\mathbf{VLM_{ij}})_{\mathbf{l\times n}} = \begin{cases} P_{Li} i = j, i \in [1,l], j \in [1,n] \\ \sum_{S_{i\to j}} \prod_{N_{i\to j}} \rho_{jout} \times P_{j\to i} i \neq j, i \in [1,l], j \in [1,n] \end{cases} \quad (2)$$

where **VGM** is a $m \times n$ matrix, **VLM** is a $l \times n$ matrix. $i \to j$ represents downstream direction from node i to node j, while $j \to i$ represents upstream direction. $S_{i\to j}$ is the set of effective paths from node i to node j, and $N_{i\to j}$ is the set of nodes included in that effective path. $P_{i\to j}$ and $P_{j\to i}$ respectively represent the input and output flow at node j in the effective path. P_{Gi} and P_{Li} represent the power of G_i and L_i at node i respectively. ρ_{jin} and ρ_{jout} are input and output ratios at node j, which are equal to the ratio of the input and output flow to the total input or total output flow at that node respectively.

2.2.2. Step 2: Calculate the virtual contribution proportion matrix for generators and loads G −VCPM/L −VCPM

Using **VGM**/**VLM** in Step 1, the virtual contribution ratio matrix **G −VCPM**/**L −VCPM** is constructed to quantitatively reflect the contribution ratios of each generator and load to each node. According to the Kirchhoff law of conservation, the input power flow must equal the output power flow for each node, so **VGM**/**VLM** can be expressed as follows:

$$\mathbf{G-VCPM} = (\mathbf{G-VCPM_{ij}})_{\mathbf{m\times n}} = \mathbf{VGM_{ij}} / \sum_{\mathbf{j=1}}^{\mathbf{n}} \mathbf{VGM_{ij}} \quad (3)$$

$$\mathbf{L-VCPM} = (\mathbf{L-VCPM_{ij}})_{\mathbf{l\times n}} = \mathbf{VLM_{ij}} / \sum_{\mathbf{j=1}}^{\mathbf{n}} \mathbf{VLM_{ij}} \quad (4)$$

2.2.3. Step 3: Calculate the virtual occupancy matrix for generators and loads G −VOM/L −VOM

The **G −VCPM**/**L −VCPM** in step 2 is used not only to determine the virtual contribution of each generator and load to each node but also to calculate the virtual occupancy of each subject on each branch. To accomplish this, a virtual occupancy matrix **G −VOM**/**L −VOM** is calculated to quantify the utilization degree of each branch by generators and loads. **G −VOM** is calculated by performing downstream tracking for all generators. The power flow of each branch is then multiplied by the virtual contribution ratio of the corresponding branch start node. Similarly, upstream tracking is employed to calculate **L −VOM** by multiplying the input power flow of each branch by the virtual contribution ratio of the start node corresponding to the branch. Specifically, the formula is:

$$\begin{cases} \mathbf{G-VOM} = (\mathbf{G-VOM_{ij}})_{\mathbf{m\times k}} = \mathbf{G-VCPM[L,J]} \times diag(P_{j,out}) \\ \mathbf{G-VCPM[L,J]} = (\mathbf{G-VCPM_{ij}})_{i\in L, j\in J}, L \in \cup_{p=1}^{m} \binom{[m]}{p}, J \in \cup_{q=s_1[1]}^{s_1[k]} \binom{[s_1[k]]}{q} \end{cases} \quad (5)$$

$$\begin{cases} \mathbf{L-VOM} = (\mathbf{L-VOM_{ij}})_{\mathbf{l\times k}} = \mathbf{L-VCPM[L,J]} \times diag(P_{j,in}) \\ \mathbf{L-VCPM[L,J]} = (\mathbf{L-VCPM_{ij}})_{i\in L, j\in J}, L \in \cup_{p=1}^{l} \binom{[l]}{p}, J \in \cup_{q=s_2[1]}^{s_2[k]} \binom{[s_2[k]]}{q} \end{cases} \quad (6)$$

where **G −VOM** is a $m \times k$ matrix, **L −VOM** is a $l \times k$ matrix. **G −VCPM**[**L**, **J**] is a submatrix of **G −VCPM**, with a size of $m \times k$. **L −VCPM**[**L**, **J**] is a submatrix of **L −VCPM**, with a size of $l \times k$. $diag(\cdot)$ is a diagonal matrix. $\cup$ represents the union operation. $P_{j,in}$ and $P_{j,out}$ represent the input and output power of branch j respectively. Let s_1 and s_2 both denote vectors whose j_{th} element $s_1[j]$ and $s_2[j]$ represent the start node and end node in branch j respectively.

2.2.4. Step 4: Calculate the virtual occupancy proportion matrix for generators and loads G −VOPM/L −VOPM

By using the **VOM** in Step 3, the virtual occupancy proportion matrix **G −VOPM**/**L −VOPM** can be calculated to quantitatively reflect the contribution of each generator and load to each branch. The contribution of all members is taken into account for each branch, which equates to the power flow flowing through it, that is, the algebraic sum of the column elements of **VOM**. Therefore:

$$\mathbf{G-VOPM}=\left(\mathbf{G-VOPM_{ij}}\right)_{m\times k}=\mathbf{G-VOM_{ij}}/\sum_{j=1}^{n}\mathbf{G-VOM_{ij}} \tag{7}$$

$$\mathbf{L-VOPM}=\left(\mathbf{L-VOPM_{ij}}\right)_{l\times k}=\mathbf{L-VOM_{ij}}/\sum_{j=1}^{n}\mathbf{L-VOM_{ij}} \tag{8}$$

2.2.5. Step 5: Load-side and generation-side network loss allocation

After analyzing the previous method, it becomes evident that network loss is allocated solely based on either the load-side or generation-side, without taking into account the impacts of both parties on the network loss, whether through downstream or upstream tracking. This result in an unfair and reasonable allocation of power loss. To address this issue, we propose a flexible and adjustable approach for bidirectional network loss allocation. Our method involves introducing an adjustable parameter, denoted as β, which can be adjusted between 0 and 1, depending on the generator, load. In line with the principle of "shared use, shared responsibility", the generator and consumer of electricity share the network loss, with $\beta = 0.5$ being allocated to each party. Specially, a power loss of β is allocated to the generator-side, while $(1-\beta)$ is allocated to the load-side.

Load-side and generation-side losses are shared as follows:

$$\mathbf{P_{LOSS_G}}=\mathbf{G-VOPM}\times diag(\mathbf{P_{LOSS}})\times\beta \tag{9}$$

$$\mathbf{P_{LOSS_L}}=\mathbf{L-VOPM}\times diag(\mathbf{P_{LOSS}})\times(1-\beta) \tag{10}$$

where $\mathbf{P_{LOSS}}$ is the branch loss matrix with the size of $k\times k$. $\mathbf{P_{LOSS_G}}$ is the power loss allocation matrix on the generation-side, with a size of $m\times k$, and $\mathbf{P_{LOSS_L}}$ is the power loss allocation matrix on the load-side, with a size of $l\times k$.

The proposed VCT-based bidirectional LA method for distribution networks is shown in Fig. 2. Using VCT, the branch losses are allocated fairly by calculating virtual contribution ratios from two-way power flow tracking. The method involves following the steps in the blue area to use downstream tracking to calculate matrices **VGM**, **G −VCPM**, **G −VOM**, and **G −VOPM** corresponding to formulas (1), (3), (5) and (7) respectively, to allocate power loss on the generation side. Moreover, the steps in the green area are followed to use upstream tracking to calculate **VLM**, **L −VCPM**, **L −VOM**, and **L −VOPM** matrices based on formulas (2), (4), (6), and (8), respectively, to determine the network loss at the load side.

2.3. Complexity analysis

In this section, we analyze the computational complexity of the LA method proposed in the previous section. To begin with, the **VGM** and **VLM** matrix in step 1, calculated using (1) and (2), require computational complexity of $O(mn)$ and $O(ln)$ respectively. Step 2 involves analyzing the **G −VCPM** and **L −VCPM** matrices obtained from (3) and (4), resulting in a computational complexity of $O(n)$. In Step 3, the main calculations involve the **G −VOM** and **L −VOM** matrices, calculated using in (5) and (6). As **G −VOM** represents m generators contributing to

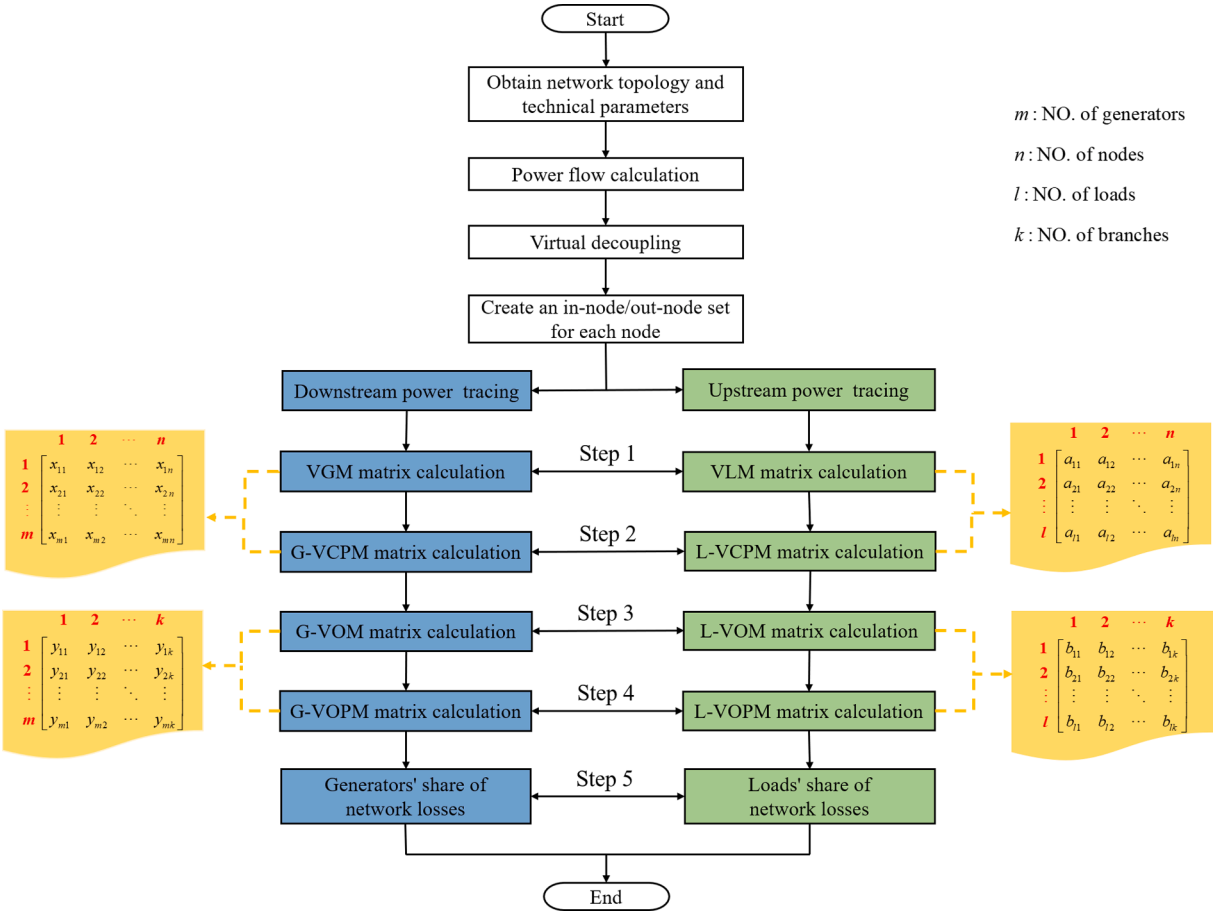


Fig. 2. Flowchart for the proposed VCT-based bidirectional power loss allocation method.

k branches, this matrix has a computational complexity of $O(mk)$. The **L** −**VOM** matrix, representing the contribution degree of l loads to k branches, has a computational complexity of $O(lk)$. Therefore, the overall computational complexity in Step 3 is about $O(mk + lk)$. In Step 4, the **G** −**VOPM** and **L** −**VOPM** matrices are calculated according to (7) and (8) based on both the **G** −**VOM** and **L** −**VOM** matrices from Step 3. Since we need to visit **G** −**VOM** and **L** −**VOM** column by column for correlation operations, the corresponding computational complexity for **G** −**VOPM** and **L** −**VOPM** matrix is $O(k)$. Finally, the network loss allocation on the generator and load sides should be calculated using (9) and (10). This step has a computational complexity of $O(k^2m)$ and $O(k^2l)$ since only two matrices are multiplied. Therefore, Step 4 has a computational complexity of $O(k^2(m + l))$. In summary, the overall computational complexity of the proposed VCT-based method is approximately $O((k^2 + n)(m + l))$.

However, the traditional PFT method has significant computational costs, mainly in solving the inverse of the upstream/downstream distribution matrices and calculating the power loss allocation on the generation-side and load-side. A downstream/upstream distribution matrix has a computational complexity of $O(n^2)$, and its inverse operation requires a computational complexity of $O(n^3)$. The calculation complexities for loss allocation on the load-side and generation-side are $O(n^3mk)$ and $O(n^3lk)$ respectively. Thus, the overall computational complexity of the PFT method is about $O(n^3k(m + l))$. For sufficiently large radial DN, $n \approx k$, where n is the number of nodes and k is the number of branches. Therefore, the total computational complexity of the proposed VCT-based method and the PFT method can be roughly deduced as $O(n^2 + n)$ and $O(n^4)$, respectively. It is evident that the proposed VCT-based method outperforms the PFT method in terms of computational efficiency.

3. Numerical case study

This section presents the verification and analysis of the proposed method using the IEEE 33 [48] and IEEE 69 [12] system to further demonstrate its benefits. Two scenarios are discussed in detail, one without and the other with DGs access, in order to analyze the impact on distribution results and network loss.

3.1. Power loss allocation study

As shown in Fig. 3, DGs with different capacities [48] are connected to nodes 14, 24, and 30. In this study, DGs are modeled as negative loads and included in the distribution load calculation, and it is assumed that the voltage magnitude and phase angle of DGs remain constant regardless of the power injection or absorption. This system operates at 12.66 kV, with load active power and reactive power of 4750 kW and 2300kVAr respectively, and an active power loss of 202.67 kW. A detailed analysis of the LA results based on two scenarios, with and without DGs, can be found in Table 1 and Fig. 4.

In Fig. 4(a), the contribution of each generator and load to each branch is illustrated, along with the distribution of the system network loss without DGs. Both the load side and the power side are responsible for half the network loss. The contribution of each load and generator to each branch varies according to its network topology and power flow distribution, and the contribution corresponds to the gradient color scale in ascending order (light blue to dark blue color). There is a significant contribution from the root node in every branch, with a contribution ratio of 100%. This is because the only power source in the system is located at this node, and power flows through all branches to power each load simultaneously. As a result, the root node power supply carries a network loss of 101.3386 kW, accounting for half of the total apportioned loss.

Fig. 4(b) illustrates how the contribute of each generator and load to each branch changes with the addition of DG. The presence of DG has caused some branches to have reversed power flow, resulting in changes to the contribution of each power source. Furthermore, the total network loss is reduced from 202.67 kW without DGs to 19.53 kW with DGs. The apportionment results from this method provide a quantitative measure of the contribution of each subject to each branch of the system, and effectively allocate power losses and enforce relevant responsibilities. Further, the loss allocation results from both scenarios satisfy the system power balance. These findings clearly demonstrate the effectiveness of the proposed method.

As shown in Table 1, there are no negative value in the network loss sharing results. It is expected that all power transactions between generators and loads will result in losses on the supply paths, as found in [27]. Science the DGs are connected to the grid at nodes 14, 24, and 30, respectively, and prioritize their local loads(L_{13},L_{23} and L_{29}), both the proposed method and the traditional method [27] have zero network loss shared by these loads. However, the method in [12] does not allocate zero network losses to local load L_{29}. This may be because the method assigns power based on “generator-load pairs”.

3.2. Computational complexity study

The Python tool is utilized in this section to compare the computational efficiency of different methods. The tool is run on a computer with an Inter-Core i7-9700 CPU, 8 GB RAM, 64-bit and Windows 10 operating system. To evaluate the performance of the proposed method, we used the IEEE 33 (in Fig. 3) and IEEE 69 (specific network topologies in [12]) systems. The results of the comparison are presented in Fig. 5. In addition, we also plotted the calculation speed curves for different methods, as shown in Fig. 6.

Using the IEEE 33 system without DGs, the proposed method demonstrates 8.7 times higher calculation efficiency, with a reduction in calculation time from 0.3825 s to 0.0439 s, and a reduction in operating memory from 792 KB to 720 KB. Similarly, a study of IEEE 69 systems under the same scenario indicates that the proposed method is 40 times more efficient than traditional methods, with the calculation time reduced from 3.3601 s to 0.0837 s. With such a large gap, the proposed

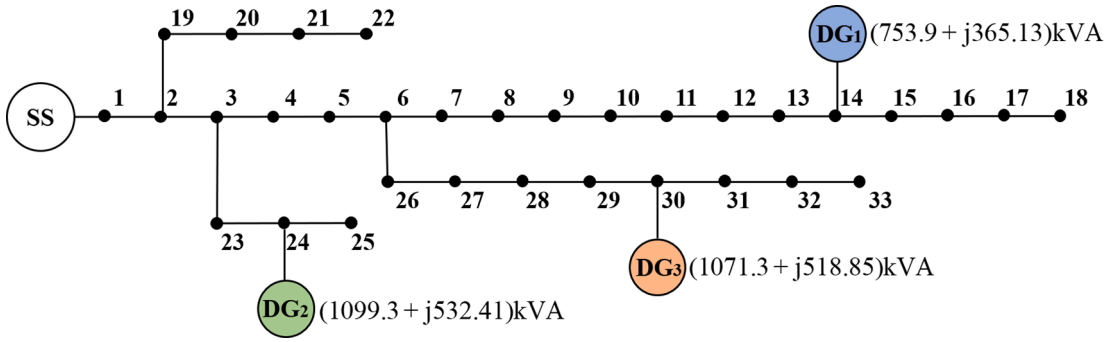


Fig. 3. Modified topology of the IEEE 33 system with multiple DGs.

Table 1
Comparison of power loss allocation results (kW) for different members (loads and DGs) in modified IEEE 33 system.

	without DGs			with DGs		
	proposed method	traditional method [27]	Literature method [12]	proposed method	traditional method [27]	Literature method [12]
L_1	0.1567	0.1567	0.1564	0.0520	0.0520	0.0449
L_2	0.8302	0.8302	0.8152	0.2235	0.2235	0.1576
L_3	1.6259	1.6259	1.6251	0.5256	0.5256	0.3867
L_4	1.0743	1.0743	1.0513	0.3826	0.3826	0.2425
L_5	1.6387	1.6387	1.6148	0.5844	0.5844	0.2538
L_6	5.6470	5.6470	5.5998	1.9957	1.9957	0.9889
L_7	6.2222	6.2222	6.1768	1.5867	1.5867	0.9160
L_8	2.0613	2.0613	2.0585	0.2212	0.2212	0.2316
L_9	2.2453	2.2453	2.2473	0.1861	0.1861	0.1955
L_{10}	1.7078	1.7078	1.6823	0.1328	0.1328	0.1278
L_{11}	2.3324	2.3324	2.3079	0.1553	0.1553	0.1525
L_{12}	2.5233	2.5233	2.4919	0.0477	0.0477	0.0463
L_{13}	5.1679	5.1679	5.0584	0.0000	0.0000	0.0000
L_{14}	2.6269	2.6269	2.6747	0.0335	0.0335	0.0364
L_{15}	2.6706	2.6706	2.6823	0.0676	0.0676	0.0681
L_{16}	2.7254	2.7254	2.7397	0.1103	0.1103	0.1131
L_{17}	4.1171	4.1171	4.0921	0.1880	0.1880	0.1819
L_{18}	0.1612	0.1612	0.1592	0.0668	0.0668	0.0585
L_{19}	0.3003	0.3003	0.2983	0.2050	0.2050	0.1967
L_{20}	0.3256	0.3256	0.3236	0.2302	0.2302	0.2218
L_{21}	0.3476	0.3476	0.3456	0.2520	0.2520	0.2436
L_{22}	0.9859	0.9859	0.9777	0.0821	0.0821	0.0813
L_{23}	5.9129	5.9129	5.8556	0.0000	0.0000	0.0000
L_{24}	6.5748	6.5748	6.5172	0.6091	0.6091	0.6091
L_{25}	1.7255	1.7255	1.6944	0.3993	0.3993	0.0936
L_{26}	1.8448	1.8448	1.7932	0.3429	0.3429	0.1053
L_{27}	2.2872	2.2872	2.1873	0.1751	0.1751	0.1418
L_{28}	5.2527	5.2527	5.0854	0.1243	0.1243	0.5702
L_{29}	9.4359	9.4359	11.2762	0.0000	0.0000	0.8033
L_{30}	7.3882	7.3882	6.9897	0.2462	0.2462	0.8179
L_{31}	10.4345	10.4345	9.8864	0.4166	0.4166	1.2354
L_{32}	2.9885	2.9885	2.8742	0.1248	0.1248	0.4456
SS	101.3386	101.3386	101.3386	3.6082	3.6082	5.5315
DG_1	–	–	–	1.4510	1.4510	1.4510
DG_2	–	–	–	1.7292	1.7292	1.6282
DG_3	–	–	–	2.9790	2.9790	1.1567
Total	202.6771	202.6771	202.6771	19.5349	19.5349	19.5349

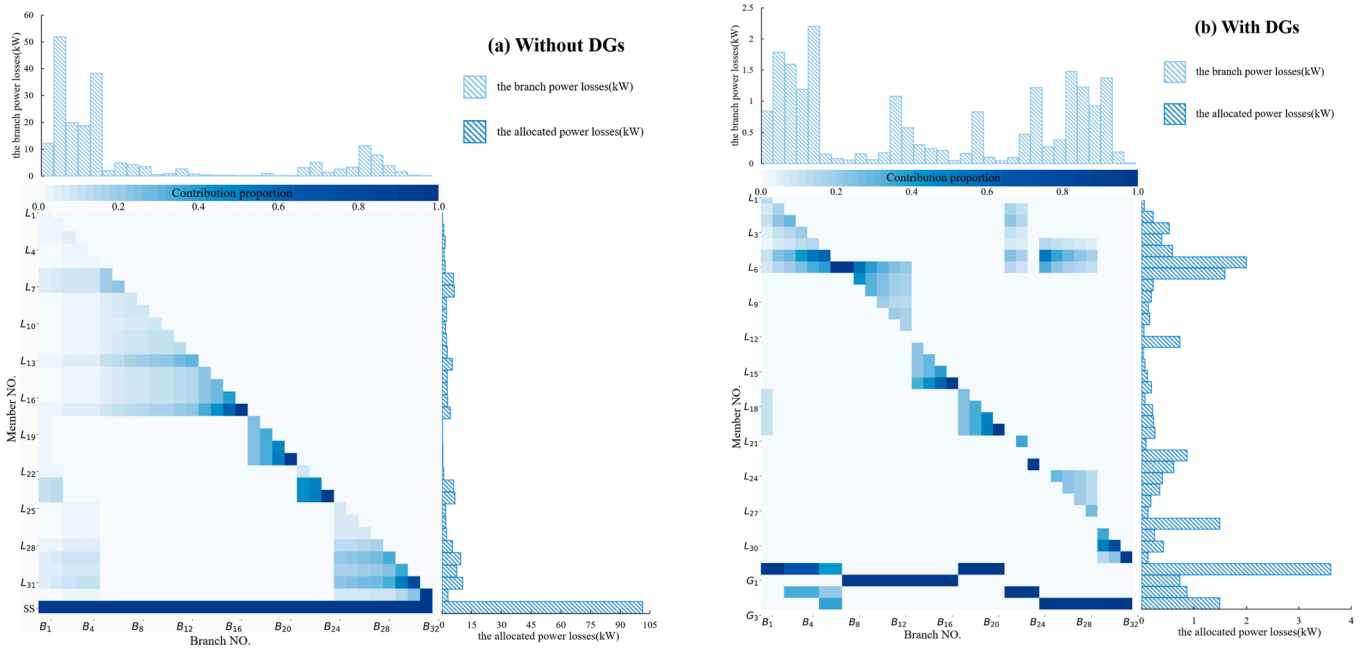


Fig. 4. The contribution proportion of generations and loads is indicated by light-to-dark colors that (a) display the results in IEEE 33 without DGs and (b) represent in that system with DGs. Different color bars show the losses allocated according to the VCT, light blue for the branch losses and dark blue for the allocated losses. The color scale shows the contribution proportion. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

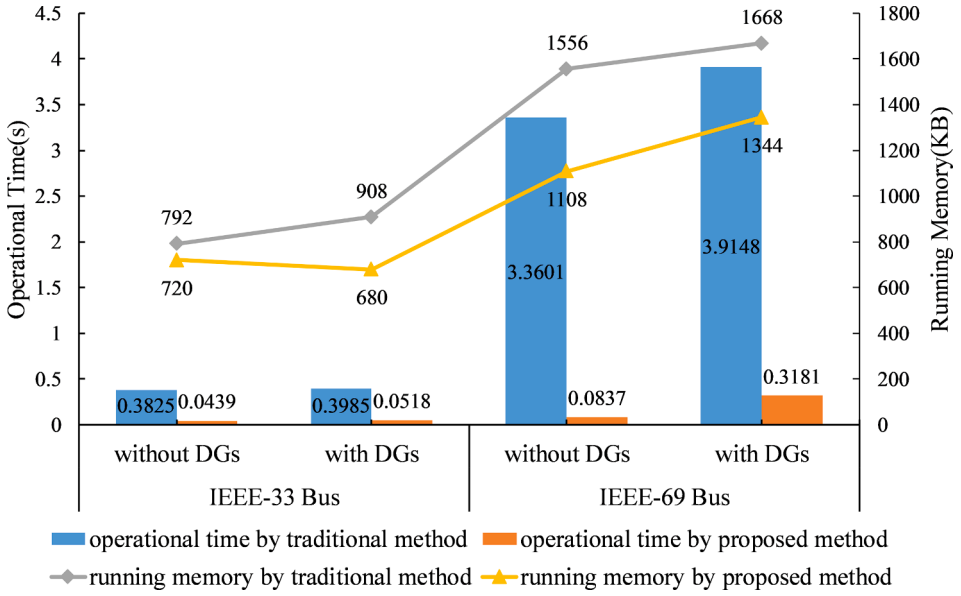


Fig. 5. Results of running time and memory comparison of different methods using the IEEE 33 and IEEE 69.

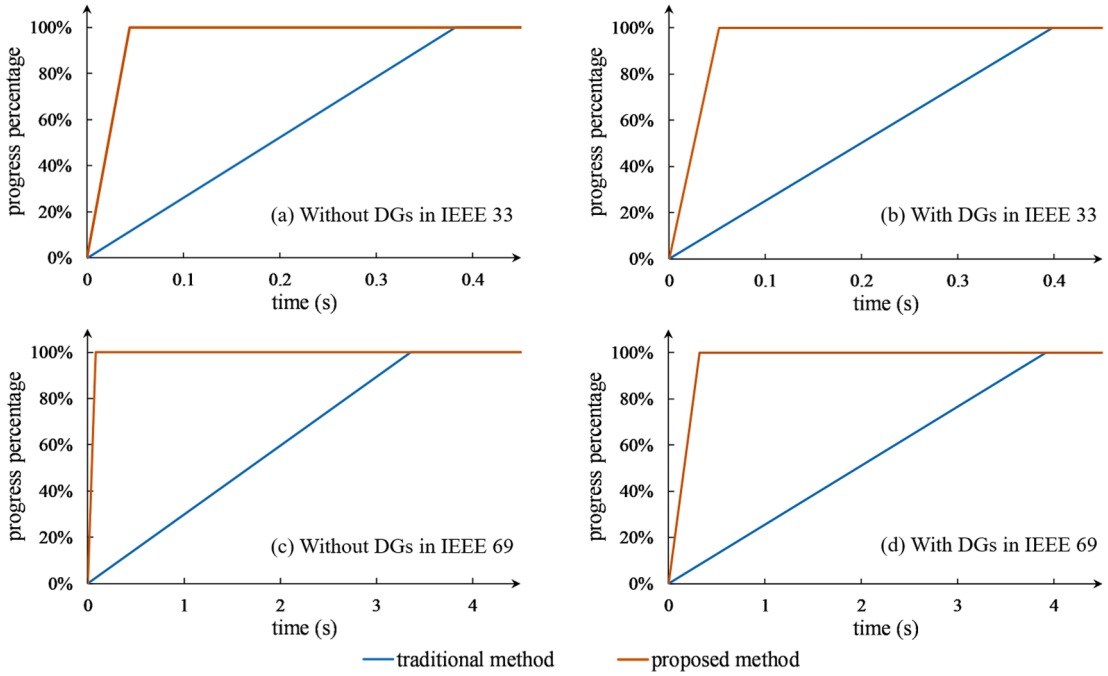


Fig. 6. Comparison of calculation speed for different methods in modified IEEE 33 and IEEE 69 system. The orange and blue lines represent the computational speed of the proposed method and traditional method [27], respectively. Panels (a) and (c) show the computational speed without DGs, while panels (b) and (d) show the computational speed with DGs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

method in this paper can greatly improve computational efficiency and reduce computational complexity. With an increasing number of system nodes and DG grid connections, computing time and running memory for the above methods will also increase. However, the calculation time of the method proposed in this paper increases by 5.14 times (from 0.0518 s to 0.3181 s), which is less than 8.82 times of the traditional method (from 0.3985 s to 3.9148 s). This is because the inverse operation of the distribution matrix in the traditional method increases computational complexity to a certain degree. This phenomenon and problem become more evident as the number of nodes increases. In addition, it also can be observed from the computational speed curve in Fig. 6 that the proposed method has a significantly faster calculation speed than the traditional method in all scenarios. In conclusion, the method proposed in this paper has a smaller computing time and running memory requirement as nodes in the test system increase, reflecting its unique ability to greatly reduce computational complexity while improving efficiency.

3.3. Results analysis

3.3.1. Accuracy assessment

In this paper, we validate the proposed VCT-based bidirectional LA method using an IEEE 33 system (shown in Fig. 3). The performance of the proposed method is compared and analyzed against the traditional

method [27] and the method in [12] using 67 sets of data from Table 1 (including 34 sets of data with DGs and 33 sets without DGs), as shown in Fig. 7.

Fig. 7(a) shows that the proposed method and the traditional method agree almost exactly with the 1:1 line, with a practically zero deviation and an average relative error of 1.49%. These results confirm the correctness, rationality and scientific validity of the proposed method. While both two methods presented here are generally consistent in terms of calculation results. However, the proposed methods have generally consistent calculation result, the proposed method is significantly less computationally complex and more efficient, as described in detail in the following section. Fig. 7(b) shows the error comparison between the proposed method and the approach presented in [12]. Almost all of the data fall within the 95% confidence interval (light red shaded area). Using the above two methods, the root mean square error (RMSE), mean absolute error (MAE) and mean relative error (MRE) are 0.467 kW, 0.204 kW and 26.01%, respectively. Additionally, we conducted a comparison between the proposed method and the literature method. The discrepancy between the results of these two methods was treated as "residuals". Following this, we employed the Durbin-Watson test to examine the autocorrelation within these "residuals". The results yielded a Durbin-Watson statistic of 1.397 between the two methods, which is close to 2, suggesting a low degree of correlation between these results, thereby signifying their independence. Additionally, since the LA results without DGs are closer to a 1:1 line and most fall within the 95% confidence interval, the error of results is smaller. It is normal for the apportionment results differ between the two LA methods due to fundamental differences in the calculations. Certainly, different allocation methods can lead to different allocation results, and there is no unique, completely correct or incorrect allocation result. Therefore, the comparison in this paper is aimed at verifying the feasibility and effectiveness of the proposed method by comparing the errors in the calculation results from different allocation methods, rather than using it as the only criterion for the correctness of allocation results. Overall, the proposed method can achieve a fair distribution of power loss between participants.

3.3.2. Computational efficiency evaluation

In this paper, we demonstrate that the proposed method has a computational complexity of $O(n^2 + n)$, which is lower than the computational complexity $O(n^4)$ in the traditional PFT method. Improving the efficiency of network loss accounting and sharing will be crucial for power distribution systems with increasingly complex and changeable topologies in the future. As illustrated in Fig. 5 and Fig. 6, the proposed method is computationally more efficient, with a speedup of 40 times compare to the traditional method, in terms of both computational time and running memory. Two main factors contribute to the efficiency of the proposed method. The first factor is the computational complexity of the method itself. In this paper, the power generation and load calculations are performed using formulas (9) and (10), which involve only multiplication between multiple sparse matrices, making the method simple and efficient. In contrast, the network loss component is calculated by inverting the distribution matrix, which increases computational complexity and can cause singularity issues. Unlike matrix inversion operations, the proposed method does not require inverse matrices, making it less computationally intensive. This aspect sets the proposed method apart from others. The second factor that makes the proposed method more efficient than traditional methods is the avoidance of computational redundancy by avoiding repeated inverse operations on the distribution matrix. The unique feature of this method is its computational complexity and operational efficiency, providing a theoretical basis and scientific guidance for future construction of new power systems as well as fine-management of network losses.

3.3.3. Applicability analysis

A virtual contribution matrix based on VCT is employed to quantify the use/occupancy of each branch by each subject system. This greatly reduces the computational complexity and improves the computational efficiency. Compared to the traditional method, the proposed method is 40 times more efficient. The high efficiency of this method is crucial for ensuring the refined and scientific management of the energy system in ADN. As the trend of grid-connected renewable energy continues to grow, multiple economic entities are demanding various information acquisitions. Therefore, it is important to consider how to achieve standardized and refined management while maintaining transparency and fairness.

The VCT-based bidirectional LA method proposed in this paper allows for the acquisition of detailed usage/occupancy information in the network without identifying individual network users. This information is accessible to all network participants, empowering them to make informed decisions. The method accurately quantifies the contribution of each branch, enabling quantitative tracking and responsibility assignment in the power system. This approach supports carbon audits, energy-saving strategies, and low-carbon power development. Furthermore, the proposed method provides a clear understanding of the relationship between power flow and transactions. It offers several advantages, including ease of understanding, low computational

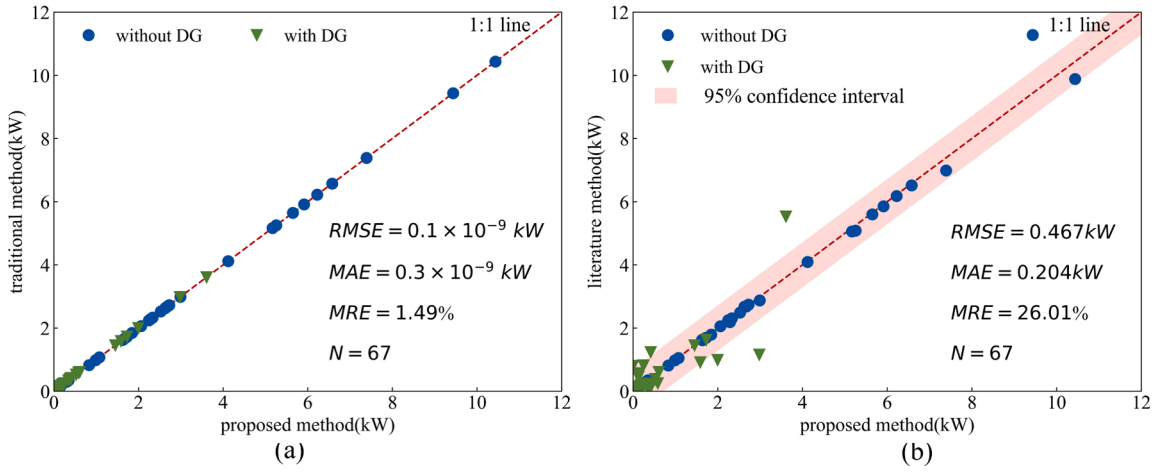


Fig. 7. Comparison of power loss allocation results among different methods in modified IEEE 33 system, plotted within a 1:1 fit line (red dotted line) and a 95% confidence interval (light red shaded area): (a) comparison between the proposed method and traditional method [27], and (b) comparison between the proposed method and literature method in [12]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

complexity, and low operation and management costs. Therefore, it has significant practical value for network cost accounting and apportionment in ADN.

4. Conclusion

This paper introduces the concept of Virtual Contribution Theory and proposes a novel bidirectional loss allocation method based on it. The method is tested and verified on IEEE 33 and IEEE 69 systems. The main conclusions are summarized as follows:

(1) using to the principle of power flow tracing, a node-based and branch-based virtual contribution matrix can be constructed to quantify the degree of utilization of the system network by the power generation and load sides. This optimizes and improves the calculation process of traditional power flow tracing, resulting in a computational complexity reduction from $O(n^4)$ to $O(n^2 + n)$. The proposed method is 40 times more efficient than the traditional method, which is essential for refined and scientific management in the new distribution network energy system.
(2) By using the bidirectional power flow tracing method based on Virtual Contribution Theory, the contributions of power losses from both the load side and generation side can be accurately quantified. Based on the principle of power distribution, network losses can be allocated according to the contribution of the generator and the user. Results show that the proposed method has a low average relative error of just 1.49%, improving the accuracy and adaptability of power loss quantitative allocation.
(3) The proposed method is an innovative bidirectional loss allocation method based on Virtual Contribution Theory. It quantifies the degree of usage and occupancy at each branch and demonstrates how each transaction pertains to the distribution of power. This method has a low computational complexity and high computational efficiency, which contributes to the refined management of energy systems. This work provides novel technical ideas and paths for accounting and allocating costs in power systems, tracking and assigning responsibilities for electricity-carbon joint, which have substantial theoretical and practical implications.

In this study, we only focused on the static allocation of losses at a specific moment. Further research is necessary to investigate the impact of network flow changes at different times on loss allocation, in order to comprehensively reveal the dynamic changes in distribution network losses. Additionally, it is worthwhile to explore dynamic loss allocation methods for different types of distributed energy resources, such as electric vehicles and energy storage devices, to enhance the operational efficiency and economic performance of distribution networks.

CRediT authorship contribution statement

Jing Zhao: Conceptualization, Methodology, Writing – original draft. **Songhuai Du:** Writing – review & editing. **Yanjun Dong:** Data curation, Software. **Juan Su:** Supervision, Writing – review & editing. **Yue Xia:** Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

This research has been funded by the Multi-timescale Forecast Technology for Large-scale Wind/Photovoltaic Power Supply Capability under the National Key Research and Development Program of China (Project Number: 2022YFB2403000).

References

[1] Caballero-Peña J, Cadena-Zarate C, Parrado-Duque A, Osma-Pinto G. Distributed energy resources on distribution networks: A systematic review of modelling, simulation, metrics, and impacts. Int J Electr Power Energy Syst 2022;138:107900. https://doi.org/10.1016/j.ijepes.2021.107900.
[2] Wu C, Gu W, Zhou S, Chen X. Coordinated Optimal Power Flow for Integrated Active Distribution Network and Virtual Power Plants Using Decentralized Algorithm. IEEE Trans Power Syst 2021;36:3541–51. https://doi.org/10.1109/TPWRS.2021.3049418.
[3] Dalhues S, Zhou Y, Pohl O, Rewald F, Erlemeyer F, Schmid D, et al. Research and practice of flexibility in distribution systems: A review. CSEE J Power Energy Syst 2019;5:285–94. https://doi.org/10.17775/CSEEJPES.2019.00170.
[4] Mohammadi A, Safdarian F, Mehrtash M, Kargarian A, A System of Systems Engineering Framework for Modern Power System Operation. In: Amini MH, Boroojeni KG, Iyengar SS, Pardalos PM, Blaabjerg F, Madni AM, editors. Sustainable interdependent networks II: From Smart Power Grids to Intelligent Transportation Networks. Springer International Publishing Ag, Cham; 2019: pp. 217–247. https://doi.org/10.1007/978-3-319-98923-5_12.
[5] Moret F, Tosatto A, Baroche T, Pinson P. Loss Allocation in Joint Transmission and Distribution Peer-to-Peer Markets. IEEE Trans Power Syst 2021;36:1833–42. https://doi.org/10.1109/TPWRS.2020.3025391.
[6] Kim J, Dvorkin Y. A P2P-Dominant Distribution System Architecture. IEEE Trans Power Syst 2020;35:2716–25. https://doi.org/10.1109/TPWRS.2019.2961330.
[7] Fairley P. The unruly power grid. IEEE Spectr 2004;41:22–7. https://doi.org/10.1109/MSPEC.2004.1318179.
[8] Wang C, Ju P, Wu F, Pan X, Wang Z. A systematic review on power system resilience from the perspective of generation, network, and load. Renew Sustain Energy Rev 2022;167:112567. https://doi.org/10.1016/j.rser.2022.112567.
[9] Sekizaki S, Nishizaki I, Hayashida T. Cooperative voltage management by demand resources and fair payoff allocation for distribution systems. Int J Electr Power Energy Syst 2023;145:108696. https://doi.org/10.1016/j.ijepes.2022.108696.
[10] Emarati M, Barani M, Farahmand H, Aghaei J, del Granado PC. A two-level over-voltage control strategy in distribution networks with high PV penetration. Int J Electr Power Energy Syst 2021;130:106763. https://doi.org/10.1016/j.ijepes.2021.106763.
[11] Songhuai Du, Xinghua Z, Mo Lu, Hui X. A novel nucleolus-based loss allocation method in bilateral electricity markets. IEEE Trans Power Syst 2006;21:28–33.
[12] Todorovski M, Rajicic D. Contribution of generator-load pairs in distribution networks power losses. Int J Electr Power Energy Syst 2020;115:105433. https://doi.org/10.1016/j.ijepes.2019.105433.
[13] Chen YC, Dhople SV. Tracing Power With Circuit Theory. IEEE Trans Smart Grid 2020;11:138–47. https://doi.org/10.1109/TSG.2019.2918258.
[14] Hörsch J, Schäfer M, Becker S, Schramm S, Greiner M. Flow tracing as a tool set for the analysis of networked large-scale renewable electricity systems. Int J Electr Power Energy Syst 2018;96:390–7. https://doi.org/10.1016/j.ijepes.2017.10.024.
[15] Song M, Cai Y, Gao C, Chen T, Yao Y, Ming H. Transactive energy in power distribution systems: Paving the path towards cyber-physical-social system. Int J Electr Power Energy Syst 2022;142:108289. https://doi.org/10.1016/j.ijepes.2022.108289.
[16] Zhao J, Tian Z, Ji H, Ji J, Yan J, Wu J, et al. Peer-to-Peer electricity trading of interconnected flexible distribution networks based on Non-Cooperative games. Int J Electr Power Energy Syst 2023;145:108648. https://doi.org/10.1016/j.ijepes.2022.108648.
[17] Wang T, Hua H, Wei Z, Cao J. Challenges of blockchain in new generation energy systems and future outlooks. Int J Electr Power Energy Syst 2022;135:107499. https://doi.org/10.1016/j.ijepes.2021.107499.
[18] Hasankhani A, Mehdi Hakimi S, Bisheh-Niasar M, Shafie-khah M, Asadolahi H. Blockchain technology in the future smart grids: A comprehensive review and frameworks. Int J Electr Power Energy Syst 2021;129:106811. https://doi.org/10.1016/j.ijepes.2021.106811.
[19] Abhyankar AR, Soman SA, Khaparde SA. Optimization approach to real power tracing: an application to transmission fixed cost allocation. IEEE Trans Power Syst 2006;21:1350–61. https://doi.org/10.1109/TPWRS.2006.879278.
[20] Chen H, Ding T, Chen L, Shi J. A modified harmonic pricing scheme for customers based on quantifying the harmonic comprehensive contribution. Int J Electr Power Energy Syst 2021;130:106905. https://doi.org/10.1016/j.ijepes.2021.106905.
[21] Hernandez JC, Ruiz-Rodriguez FJ, Jurado F, Sanchez-Sutil F. Tracing harmonic distortion and voltage unbalance in secondary radial distribution networks with photovoltaic uncertainties by an iterative multiphase harmonic load flow. Electr Pow Syst Res 2020;185:106342. https://doi.org/10.1016/j.epsr.2020.106342.
[22] Wang S, Liu X, Wang K, Wu L, Zhang Y. Tracing harmonic contributions of multiple distributed generations in distribution systems with uncertainty. Int J Electr Power Energy Syst 2018;95:585–91. https://doi.org/10.1016/j.ijepes.2017.09.014.
[23] Cvijić S, Ilić MD. Part I: A New Framework for Modeling and Tracing of Bilateral Transactions and the Corresponding Loop Flows in Multi-Control Area Power

Networks. IEEE Trans Power Syst 2014;29:2706–14. https://doi.org/10.1109/TPWRS.2014.2312376.
[24] Enshaee A, Yousefi GR, Ebrahimi A. Allocation of transmission active losses through a novel power tracing-based technique. IET Gener Transm Distrib 2018; 12:3201–11. https://doi.org/10.1049/iet-gtd.2017.1519.
[25] Rouholamini M, Miller CJ, Wang C. Determining consumer's carbon emission obligation through virtual emission tracing in power systems. Environ Prog Sustain Energy 2020;39:13279. https://doi.org/10.1002/ep.13279.
[26] Wang C, Wang Y, Rouholamini M, Miller C. An Equivalent Circuit-Based Approach for Power and Emission Tracing in Power Networks. IEEE Syst J 2022;16:2206–16. https://doi.org/10.1109/JSYST.2021.3067296.
[27] Achayuthakan C, Dent CJ, Bialek JW, Ongsakul W. Electricity Tracing in Systems With and Without Circulating Flows: Physical Insights and Mathematical Proofs. IEEE Trans Power Syst 2010;25:1078–87. https://doi.org/10.1109/TPWRS.2009.2037506.
[28] Bialek JW, Kattuman PA. Proportional sharing assumption in tracing methodology. IEE Proc - Gener, Trans Distrib 2004;151:526–32. https://doi.org/10.1049/ip-gtd:20040351.
[29] Kiani-Moghaddam M, Shivaie M, Arabkoohsar A. Towards a new effective strategy to obtain optimal radial structure in power distribution networks: Graph theory-based topology assessment. Int J Electr Power Energy Syst 2022;143:108484. https://doi.org/10.1016/j.ijepes.2022.108484.
[30] Nikolaidis AI, Charalambous CA, Mancarella P. A Graph-Based Loss Allocation Framework for Transactive Energy Markets in Unbalanced Radial Distribution Networks. IEEE Trans Power Syst 2019;34:4109–18. https://doi.org/10.1109/TPWRS.2018.2832164.
[31] Bharti D, De M. A new graph theory based loss allocation framework for bilateral power market using diakoptics. Int J Electr Power Energy Syst 2016;77:395–403. https://doi.org/10.1016/j.ijepes.2015.11.014.
[32] Gubina F, Grgic D, Banic I. A method for determining the generators' share in a consumer load. IEEE Trans Power Syst 2000;15:1376–81. https://doi.org/10.1109/59.898115.
[33] Wu T, Hu B, Xie K, Tai H-M, Wang S, Li B. A power flow tracing based load curtailment technique, in. IEEE Innov Smart Grid Technol - Asia (ISGT Asia) 2019; 2019:3855–9. https://doi.org/10.1109/ISGT-Asia.2019.8881164.
[34] Al-Digs A, Chen YC. Power System Loss Divider. IEEE Trans Power Syst 2020;35: 3286–9. https://doi.org/10.1109/TPWRS.2020.2989632.
[35] Khalid SN, Shareef H, Mustafa MW, Khairuddin A, Maungthan Oo A. Evaluation of real power and loss contributions for deregulated environment. Int J Electr Power Energy Syst 2012;38:63–71. https://doi.org/10.1016/j.ijepes.2011.12.021.
[36] Rajičić D, Todorovski M. Participation of Every Generator to Loads, Currents, and Power Losses. IEEE Trans Power Syst 2021;36:1638–40. https://doi.org/10.1109/TPWRS.2020.3044485.
[37] Enshaee A, Enshaee P. New reactive power flow tracing and loss allocation algorithms for power grids using matrix calculation. Int J Electr Power Energy Syst 2017;87:89–98.
[38] Narimani M, Hosseinian SH, Vahidi B. A modified methodology in electricity tracing problems based on Bialek's method. Int J Electr Power Energy Syst 2014; 60:74–81. https://doi.org/10.1016/j.ijepes.2014.02.011.
[39] Sindi H, Nour M, Rawa M, Öztürk Ş, Polat K. Random fully connected layered 1D CNN for solving the Z-bus loss allocation problem. Measurement 2021;171: 108794.
[40] Sulaiman MH, Mustafa MW, Shareef H, Abd SN. Khalid, An application of artificial bee colony algorithm with least squares support vector machine for real and reactive power tracing in deregulated power system. Int J Electr Power Energy Syst 2012;37:67–77. https://doi.org/10.1016/j.ijepes.2011.12.007.
[41] Mustafa MW, Sulaiman MH, Shareef H, Khalid SNA. Reactive power tracing in pool-based power system utilising the hybrid genetic algorithm and least squares support vector machine. IET Gener Transm Distrib 2012;6:133–41. https://doi.org/10.1049/iet-gtd.2011.0166.
[42] Usman M, Coppo M, Bignucolo F, Turri R. Losses management strategies in active distribution networks: A review. Electr Pow Syst Res 2018;163:116–32. https://doi.org/10.1016/j.epsr.2018.06.005.
[43] Haque MM, Wolfs P. A review of high PV penetrations in LV distribution networks: Present status, impacts and mitigation measures. Renew Sustain Energy Rev 2016; 62:1195–208. https://doi.org/10.1016/j.rser.2016.04.025.
[44] Karimi M, Mokhlis H, Naidu K, Uddin S, Bakar AHA. Photovoltaic penetration issues and impacts in distribution network – A review. Renew Sustain Energy Rev 2016;53:594–605. https://doi.org/10.1016/j.rser.2015.08.042.
[45] Mirhassani S, Ong HC, Chong WT, Leong KY. Advances and challenges in grid tied photovoltaic systems. Renew Sustain Energy Rev 2015;49:121–31. https://doi.org/10.1016/j.rser.2015.04.064.
[46] Ghofrani-Jahromi Z, Mahmoodzadeh Z, Ehsan M. Distribution Loss Allocation for Radial Systems Including DGs. IEEE Trans Power Delivery 2014;29:72–80. https://doi.org/10.1109/TPWRD.2013.2277717.
[47] Jagtap KM, Khatod DK. Loss allocation in radial distribution networks with various distributed generation and load models. Int J Electr Power Energy Syst 2016;75: 173–86. https://doi.org/10.1016/j.ijepes.2015.07.042.
[48] Kumar P, Gupta N, Niazi KR, Swarnkar A. Branch current decomposition method for loss allocation in contemporary distribution systems. Int J Electr Power Energy Syst 2018;99:134–45.